0. Course Presentation

Operating Systems (OS)
Master in Innovation and Research in Informatics (MIRI)
Facultat d'Informàtica de Barcelona (FIB)
Universitat Politècnica de Catalunya (UPC)
2025/2026 Q1

FIB

Instructor

- Instructor: Jordi Guitart
- E-mail: jordi.guitart@upc.edu
- Office: C6-205
- Office Hours: Arrange an appointment by mail to meet physically or virtually (e.g. Google Meet)

Instructor

- Instructor: Juan José Costa
- E-mail: juan.jose.costa@upc.edu
- Office: C6-117
- Office Hours: Arrange an appointment by mail to meet physically or virtually (e.g. Google Meet)

Goals

- Understand the definition of an O.S. for a warehouse-scale computer, as well as the challenges for its design and implementation
- Understand the features of an O.S. for a warehouse-scale computer, whether they evolve from a desktop computer O.S. or they are brand-new
- Know how some features of an O.S. for a warehouse-scale computer are implemented in real systems

Contents

1. Desktop Computers: Architecture & O.S. Review
2. Warehouse-scale Computers: Features & Challenges
3. Process Management
4. In-memory Data Management
5. On-disk-storage Data Management
6. Energy Management

Course organization

- 10 h/week of student work

1. Guided learning (4 h/week)
   - **A. Lecture classes**
     - Objective: Acquire theoretical knowledge
   - **B. Practical seminar sessions**
     - Objective: Apply in practice theoretical concepts
     - Not shown in the official timetable ⇒ See our calendar!
2. Independent learning (6 h/week)

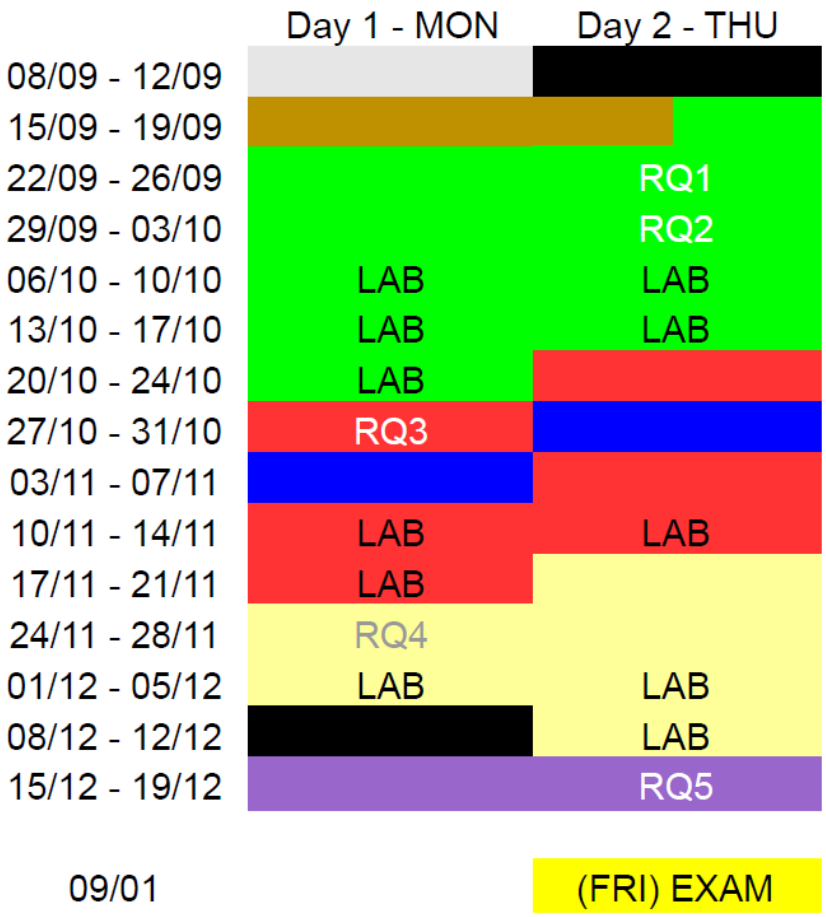
Lecture classes

- Slide-based lectures
- Exercises (from previous exams)
  - Available in the Exam documentation area in the 'Racó'
- Online quizzes: **Wayground** (formely Quizizz)
  - https://wayground.com/
  - Register if you want to keep your history of quizzes
- Preparation/Follow-up: Reading of papers
  - Answer of questionnaires asking about papers content
  - Papers content may be requested in the final exam

Practical seminar sessions

- Lab work in teams of four students
  - Answer the activity in ATENEA to fill the team members before September, 22nd EOB
    - https://atenea.upc.edu/mod/choicegroup/view.php?id=5018206
- **You need to bring your laptop!**
  - It should have (at least) two x86_64 / amd64 processors with virtualization support (VT-x or AMD-V) and (at least) 2 GiB of RAM
  - We provide an external USB hard drive to each student and an Ethernet router to each team
    - We will notify you when you can get your IT equipment in the FIB operation service located in the A5 building

Calendar



Presentation+OS Review
WSCs: Features+Challenges
Process Mgmt
In-memory Data Mgmt
On-disk-storage Data Mgmt
Energy Mgmt
Midterm period (NO classes)
Bank holidays
Change of schedule

RQi: Reading Questionnaire

Course material

- We have a workspace in **ATENEA**
  - Resources:
    - Calendar
    - Lecture slides
    - Reading assignments
    - Practical seminar assignments
    - Further reading material
    - Bibliography
  - Activities:
    - Reading questionnaires

  ⇒ https://atenea.upc.edu/course/view.php?id=99782

Grading

A. Written exam
B. Evaluation of practicals
C. Evaluation of readings

- Final Grade for OS-MIRI =

$$\mathbf{0.55} \times \boldsymbol{A} + \mathbf{0.3} \times \boldsymbol{B} + \mathbf{0.15} \times \boldsymbol{C}$$

Grading

A. Written exam

1. Multiple-choice test about the lectures (5 points)
   ⇒ Closed-book: you are not allowed notes, books or any other reference material, including electronic devices
2. Exercises about the lectures (5 points)
   ⇒ Restricted open-book: you are allowed a **single legible double-sided cheat sheet**

⇒ Confirmed date: **09/01/2026**, 11.30h

Grading

B. Practicals grade

- **Three** practical assignments during the course
- Each assignment includes:

1. Guided practice activities (8 points)
   - They are evaluated during the sessions through the accomplishment of a number of **milestones**
2. Independent practice activities (2 points)
   - You can choose from a list of activities
   - They are evaluated out-of-class by means of meetings, reports, videos, …, depending on the activities done

- The final grade for the practicals is calculated as the average grade of the three assignments

Grading

C. Readings grade

- **Five** reading assignments during the course
- One questionnaire at ATENEA for each reading
  - To be answered one week after the reading is announced (at the beginning of the class)

    ⇒ See our calendar!
  - 15 minutes to answer 6 questions + personal opinion
- The final grade for the readings is calculated as the average grade of the five questionnaires

Academic integrity

- We adhere to the commitment to academic integrity at the UPC
  - All the work you submit for grading, individually or as a team, must be your own work
  - Plagiarism and fraud are unacceptable, whether the source is a current/past student, or a website
  - Aiding someone else to cheat is also fraudulent
- Violations of academic integrity will be severely sanctioned as per the FIB protocol

Bibliography

- Basic textbooks
  - Barroso, L.A., Hölzle, U., Ranganathan, P., *The Datacenter as a Computer: An Introduction to the Design of Warehouse-Scale Machines*, 3rd edition, Morgan and Claypool Publishers, 2018
- Additional books
  - Bugnion, E., Nieh, J., Tsafrir, D., *Hardware and Software Support for Virtualization*, Morgan and Claypool Publishers, 2017
  - Silberschatz, A., Galvin, P.B., Gagne, G., *Operating System Concepts*, 10th edition, John Wiley & Sons, Inc., 2018
  - Arpaci-Dusseau, R.H., Arpaci-Dusseau, A.C., *Operating Systems: Three Easy Pieces*, v. 1.10, Arpaci-Dusseau Books, 2023
  - Tanenbaum, A., Bos, H., *Modern Operating Systems*, 5th edition, Pearson, 2023
  - Hennessy, J.L., Patterson, D.A., *Computer Architecture: A Quantitative Approach*, 6th Ed., Elsevier, Morgan Kaufmann, 2017

Requirements

- OS concepts (SO)
- Networks concepts (XC)
- Architecture concepts (AC, MP)